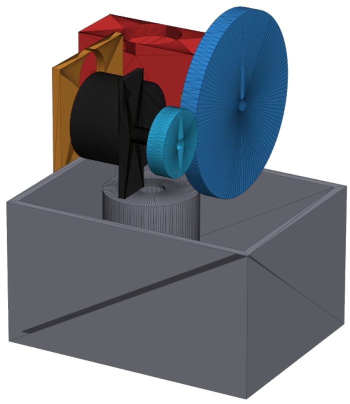
Geared Pan/Tilt Laser-Pointer Mechanism

A Beginner's Guide to the Parts, the Movement, and the Build



This guide explains, in plain language, how this small motorised mount works: it can swivel left-right (**pan**) and tip up-down (**tilt**) to aim a laser pointer or similar small device in almost any direction. No prior mechanical design or CAD experience is assumed - every technical word is explained the first time it is used.

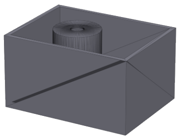
Total parts (numbered)	**18**
Approx. assembled mass	**333 g**
Pan / tilt gear reduction	**3 : 1 (each axis)**
Tilt holding-torque safety margin	**19.6x**

How this guide is organised: Chapter 1 introduces every physical part with a picture and a number. Chapters 2 and 3 refer back to those same part numbers throughout, so you can always flip back to Chapter 1 if you forget what "Part 6" is.

Chapter 1 - Parts of the Design

Every physical piece of this mechanism is listed below with a number, a picture, and a plain-language description. Each part is also labelled **3D PRINTED** (made on a home 3D printer from plastic filament) or **BOUGHT** (a ready-made part purchased online or from a hardware shop). Quantities show how many of that exact part are used in the whole mechanism.

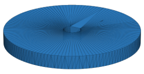
#	Part name	Type	Qty
1	Base Shell	3D Printed	1
2	Pan Gear (60 tooth)	3D Printed	1
3	Pan Motor Pinion (20 tooth)	3D Printed	1
4	Pan Platform + Tilt Walls	3D Printed	1
5	Yoke	3D Printed	1
6	Tilt Gear (60 tooth)	3D Printed	1
7	Tilt Motor Pinion (20 tooth)	3D Printed	1
8	Tilt Motor Bracket	3D Printed	1
9	Pan Shaft	Bought	1
10	Tilt Shaft	Bought	1
11	Pan Bearing 608ZZ	Bought	2
12	Tilt Bearing 625ZZ	Bought	2
13	Pan Motor (28BYJ-48 + ULN2003 driver)	Bought	1
14	Tilt Motor (28BYJ-48 + ULN2003 driver)	Bought	1
15	M4 Bolts (motor mounting)	Bought	4
16	M3 Grub Screws (shaft keying)	Bought	2
17	M3 Bolts (bracket fixings)	Bought	8
18	Laser Pointer / Payload	Bought	1



1. Base Shell

3D Printed Qty: 1

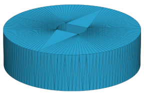
The outer box the whole mechanism is built on. Hollow inside, open on top, with a round tower in the middle that holds the two pan bearings. Bolts down to a table or tripod plate.



2. Pan Gear (60 tooth)

3D Printed Qty: 1

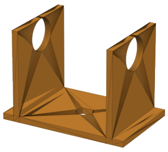
The big gear inside the base. It is locked onto the pan shaft, so when it turns, the shaft, the platform above it, and everything on the platform all turn with it. This is the 'PAN' motion.



3. Pan Motor Pinion (20 tooth)

3D Printed Qty: 1

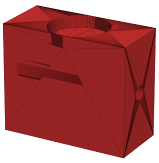
A small gear pressed onto the pan motor's own output shaft. It meshes with Part 2 (the big pan gear) and drives it. Its own position never moves - only the base motor spins it in place.



4. Pan Platform + Tilt Walls

3D Printed Qty: 1

A flat deck with two upright walls. It sits on top of the base and is keyed to the pan shaft, so it swings left-right with the pan motion. The two walls hold the tilt bearings and carry the entire tilt stage (yoke, tilt gear, tilt motor) around with them.

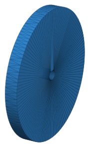


5. Yoke

3D Printed Qty: 1

The part that actually holds the laser pointer. It is locked onto the tilt shaft between the two walls, so it tips up and down for 'TILT', while also being carried left-right whenever the platform pans. It has a round pocket on top sized for the payload.

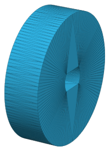
6. Tilt Gear (60 tooth)



3D Printed Qty: 1

The big gear for the tilt axis, mounted outboard of one wall. Locked to the same shaft as the yoke, so turning this gear IS the tilt motion. It rides along with pan the same way the yoke does.

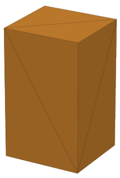
7. Tilt Motor Pinion (20 tooth)



3D Printed Qty: 1

Small gear on the tilt motor's own shaft, meshing with Part 6. It is carried around by pan (because the tilt motor is bolted to the platform) but does not itself tip up and down - it just spins in place to drive the tilt gear.

8. Tilt Motor Bracket



3D Printed Qty: 1

A small bridging plate that bolts the tilt motor to wall B of the platform. Simple structural part - travels with pan, never moves relative to the platform.

9. Pan Shaft

Bought (steel rod, cut to length) Qty: 1

A ground steel rod, 8mm diameter x ~51mm long. Printed shafts wear out of round quickly, so this one part is always metal, never printed. Filed flat on one side so a grub screw can grip it without slipping.

10. Tilt Shaft

Bought (steel rod, cut to length) Qty: 1

Same idea as Part 9, smaller: 5mm diameter x ~65mm long steel rod that the yoke and tilt gear are both keyed to.

11. Pan Bearing 608ZZ

Bought Qty: 2

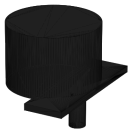
A ball bearing is a metal ring-within-a-ring with small steel balls rolling between them, so one ring can spin smoothly against the other with very little friction. Two of these (8x22x7mm) are pressed into the base tower; their outer ring stays still, their inner ring spins with the pan shaft.

12. Tilt Bearing 625ZZ

Bought Qty: 2

Same idea as Part 11, smaller (5x16x5mm). One is pressed into each platform wall and lets the tilt shaft spin smoothly. The bearing housings travel with pan; the shaft inside also tilts.

13. Pan Motor (28BYJ-48 + ULN2003 driver)



Bought Qty: 1

A small geared stepper motor. 'Stepper' means it turns in tiny, precise steps instead of spinning freely, so the electronics always know exactly which way it is pointing. Bolted directly to the base floor; its body never moves, only its shaft spins.

14. Tilt Motor (28BYJ-48 + ULN2003 driver)

Bought Qty: 1

Identical motor to Part 13, mounted instead on the platform (via Part 8) so it travels with pan while its shaft drives tilt.

15. M4 Bolts (motor mounting)

Bought Qty: 4

Generic M4 bolts, two per motor, holding Parts 13 and 14 to their mounting tabs. (Shown here as a plain generic bolt shape - the exact head style is not critical.)

16. M3 Grub Screws (shaft keying)

Bought Qty: 2

Small headless screws that tighten sideways through a printed part into the flat filed on a shaft, locking the two together so they rotate as one piece. Used to lock Part 2 to Part 9 and Part 5 to Part 10.

17. M3 Bolts (bracket fixings)

Bought Qty: 8

General-purpose M3 bolts used to fasten Part 8 to the platform wall and to close up any other printed-to-printed joints.

18. Laser Pointer / Payload

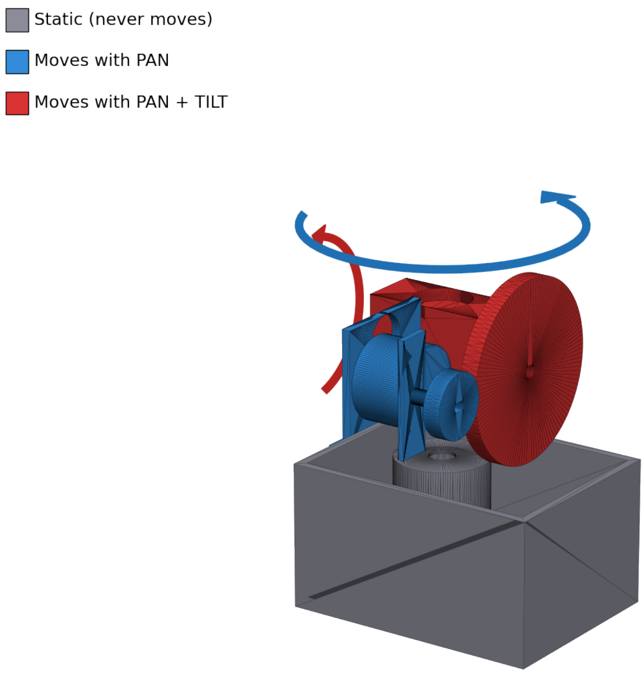
Bought (user-supplied) Qty: 1

The actual laser pointer (or other small device) the mechanism aims. Sits in the round pocket on top of the yoke (Part 5). Shown here as a generic pen-shaped placeholder, since the source design does not model a specific product - the real item will vary.

Chapter 2 - Movement

This mechanism moves in two independent directions, called axes:

- **Pan** - swivels the whole top of the mechanism left and right, like shaking your head 'no'. The entire upper assembly (Parts 2, 4, 7, 8, 9, 12, 14, and everything mounted on the platform) spins together around a single vertical axis running up through the middle of the base.
- **Tilt** - tips the pointing end up and down, like nodding your head 'yes'. Only the yoke (Part 5), the tilt gear (Part 6), and the tilt shaft (Part 10) actually rotate about this second axis - but because this second axis is carried on top of the pan stage, these parts also swing left-right whenever the mechanism pans.



Grey parts are bolted down and never move. Blue parts move only when the mechanism pans. Red parts are carried around by pan AND rotate on their own for tilt - notice that nothing in this design tilts without also being able to pan, because the tilt axis sits on top of the pan stage, not the other way around.

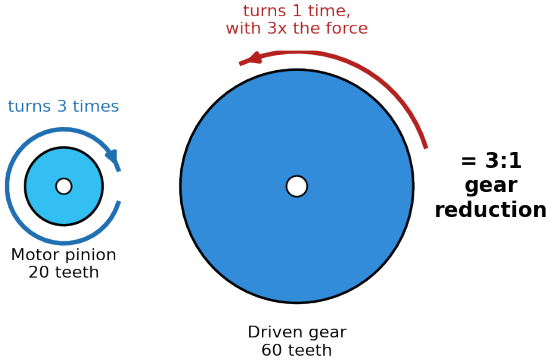
#	Part	How it moves
1	**Base Shell**	Bolted to the table/tripod. Never moves.
2	**Pan Gear (60 tooth)**	Spins in place about the vertical pan axis - this spin IS the pan motion.
3	**Pan Motor Pinion (20 tooth)**	Bolted inside the static base; spins on the spot to drive Part 2.
4	**Pan Platform + Tilt Walls**	Whole platform swings left-right with pan; carries the entire tilt stage.
5	**Yoke**	Carried around by pan; tips up/down for tilt.
6	**Tilt Gear (60 tooth)**	Carried around by pan; spinning it about the tilt axis IS the tilt motion.
7	**Tilt Motor Pinion (20 tooth)**	Carried around by pan (bolted to the platform); spins on the spot to drive Part 6.
8	**Tilt Motor Bracket**	Bolted to the platform; travels with pan only.
9	**Pan Shaft**	The steel rod Parts 2 and 4 are keyed to; physically spins to create pan.
10	**Tilt Shaft**	The steel rod Parts 5 and 6 are keyed to; carried by pan, spins for tilt.

#	Part	How it moves
11	**Pan Bearing 608ZZ**	Outer ring pressed into the static base; inner ring spins with Part 9.
12	**Tilt Bearing 625ZZ**	Outer ring pressed into the platform walls (moves with pan); inner ring spins with Part 10.
13	**Pan Motor (28BYJ-48)**	Bolted to the base floor. Only its internal shaft spins.
14	**Tilt Motor (28BYJ-48)**	Bolted to the platform via Part 8; travels with pan, its shaft drives tilt.
18	**Laser Pointer / Payload**	Sits in the pocket on Part 5; points wherever pan + tilt aim it.

How the motors turn into motion

Both axes use exactly the same trick, just aimed in different directions. Each axis has its own small motor (Part 13 for pan, Part 14 for tilt). These are **stepper motors** - a type of electric motor that rotates in tiny, fixed-size steps rather than spinning freely, so the control electronics always know exactly how far it has turned without needing a separate sensor.

Neither motor drives its axle directly. Instead, a small 20-tooth gear (a **gear** is a toothed wheel that locks its teeth into a matching wheel so the two always turn together at a fixed ratio) is fixed to the motor's own shaft, and it meshes with a much bigger 60-tooth gear fixed to the axis it is driving. This is called a **gear reduction**:



Because the big gear has 3 times as many teeth as the small one, the small gear must spin all the way round 3 times to make the big gear turn once. The trade-off is exactly reversed for turning force: the big gear turns with 3 times the force (called **torque** - a twisting force, like the effort needed to turn a stiff doorknob) that the motor produces on its own. This is why a small, cheap motor can still hold a payload steady against gravity - the gears multiply its strength at the cost of speed.

Motion path, start to finish: electrical pulses spin the pan motor (13) → its pinion (3) turns the pan gear (2) 3x slower/3x stronger → the pan gear is keyed to the pan shaft (9) → the pan shaft carries the whole platform (4) and everything bolted to it around with it. Separately, pulses spin the tilt motor (14, itself riding on the platform) → its pinion (7) turns the tilt gear (6) 3x slower/3x stronger → the tilt gear is keyed to the tilt shaft (10), which also carries the yoke (5) → the yoke tips the payload (18) up or down. Combine both motors at once and the laser can be aimed anywhere within its pan and tilt range.

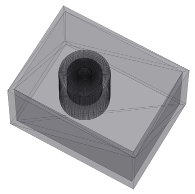
Gear reduction ratio: **3:1** on each axis (20 teeth to 60 teeth). Internally, the 28BYJ-48 motors themselves already contain their own small gearbox (about 64:1) before their output shaft even appears, so the true total reduction from the electric motor's own rotor to the final pointing direction is roughly **190:1** - this is why the mechanism can hold its position firmly with the power off and needs no brake.

Backlash note: printed gears always have a small amount of **backlash** - a tiny bit of play or looseness between meshing teeth, similar to the slack in a bicycle chain. For this design that is roughly 0.3-0.5 degrees at the output, noticeably more than the motor's own step size. In practice this means: always approach a target angle from the same direction for repeatable, accurate pointing.

Chapter 3 - Assembly

Build the mechanism from the base upward, testing that each stage spins freely by hand before adding the next. Two kinds of joints are used throughout, and each step below says which kind applies:

- **Fixed / rigid joint** - bolted tight, glued, or press-fit so the two parts never move relative to each other again.
- **Moving joint** - a bearing or a shaft spinning inside a hole, deliberately free to rotate.

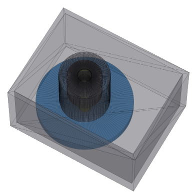


Step 1 - Prepare the base and pan bearings

• Note: Steps 1-3 show the Base Shell (Part 1) as see-through (like glass) so you can see the parts being fitted inside it. The real base is solid plastic - this is just for clarity.

• Take the Base Shell (Part 1) and press a Pan Bearing 608ZZ (Part 11, x2) into each of the two round pockets in the central tower, 16mm apart. This is a tight **press-fit** (pushed in firmly by hand or with an arbor press, no glue needed) - a **moving joint**, since the bearings' whole purpose is to let something spin smoothly inside them.

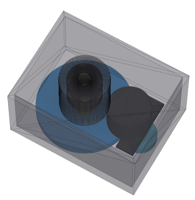
• Tip: if a bearing goes in too loosely, print a slightly larger test pocket and recalibrate (see project notes) before committing to the final print.



Step 2 - Fit the pan shaft and pan gear

• Slide the Pan Shaft (Part 9, steel rod) up through both Part 11 bearings from underneath - it should spin completely freely once through (**moving joint**).

• Slide the Pan Gear (Part 2) onto the shaft near the bottom and tighten an M3 Grub Screw (Part 16) sideways through the gear's hub onto the flat filed into the shaft. This is a **fixed joint** - the gear and shaft now always turn together.

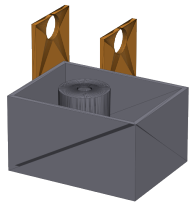


Step 3 - Add the pan motor

• Press the Pan Motor Pinion (Part 3) onto the Pan Motor's (Part 13) own output shaft - **fixed joint**, a firm press-fit onto the motor's D-shaped shaft.

• Bolt the motor down inside the base using 2x M4 Bolts (Part 15) through its mounting tabs into the printed bosses in the base floor - **fixed joint**.

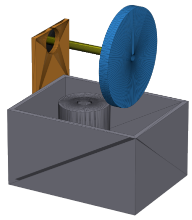
• Check the pinion (Part 3) meshes with the pan gear (Part 2) with a small amount of visible play (backlash) between the teeth, not jammed tight and not loose enough to skip.



Step 4 - Fit the pan platform

• Lower the Pan Platform + Tilt Walls (Part 4) onto the base so its central D-shaped hole slides down over the top of the pan shaft (Part 9) and rests on the base rim.

• Tighten an M3 Grub Screw (Part 16) through the platform hub onto the shaft's flat - **fixed joint**. From this point on, the entire upper structure should swing left-right by hand whenever you turn the shaft - that is the pan motion working correctly.

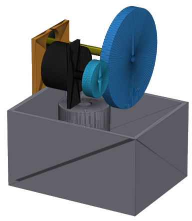


Step 5 - Fit the tilt bearings, tilt shaft, and tilt gear

• Press one Tilt Bearing 625ZZ (Part 12) into each of the two platform walls - **moving joint**, same idea as Step 1 but smaller.

• Press the Tilt Gear (Part 6) onto one end of the Tilt Shaft (Part 10, steel rod) and lock it with an M3 Grub Screw (Part 16) - **fixed joint**.

• Slide the shaft through both Part 12 bearings so the gear sits just outboard of one wall - it should spin freely.

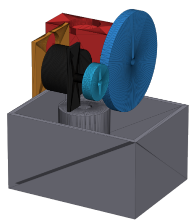


Step 6 - Add the tilt motor

• Press the Tilt Motor Pinion (Part 7) onto the Tilt Motor's (Part 14) shaft - **fixed joint**, same as Step 3.

• Bolt the Tilt Motor Bracket (Part 8) to the outer platform wall using M3 Bolts (Part 17), then bolt the motor to the bracket with 2x M4 Bolts (Part 15) - both **fixed joints**.

• Check the pinion (Part 7) meshes cleanly with the tilt gear (Part 6), same backlash check as Step 3.

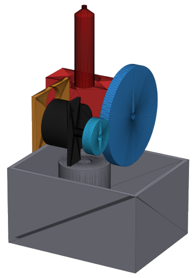


Step 7 - Fit the yoke

• Slide the Yoke (Part 5) onto the tilt shaft (Part 10) between the two platform walls, centred on the shaft.

• Tighten an M3 Grub Screw (Part 16) through the yoke hub onto the shaft's flat - **fixed joint**. The yoke, tilt shaft, and tilt gear are now permanently locked together and rotate as a single unit for tilt.

• By hand, tip the yoke through its full range and confirm it clears the platform walls at every angle (the design keeps a small but deliberate clearance here - see Part 5 notes).



Step 8 - Fit the payload

• Fit the Laser Pointer / Payload (Part 18) into the round pocket on top of the yoke (Part 5) last, once the rest of the mechanism is confirmed to move freely.

• Fitting it last lets you balance it roughly centred over the tilt axis, which keeps the holding torque the tilt motor needs as low as possible.

• Wiring note: the tilt motor's cable crosses the pan joint. Leave a service loop of slack cable, limit how far the mechanism can pan in software, or fit a slip ring at the pan axis if you need continuous, unlimited pan rotation.

This is a beginner-oriented explanation of a design-intent CAD model. Real-world dimensions (especially the exact 28BYJ-48 motor geometry and bearing pocket sizes) should be verified against your own physical parts before committing to a final print - see the project's engineering notes for the full list of flagged risks and calibration steps.